

Wastewater-based Epidemiology for Antimicrobial Resistance

Mingjun Jiang, Environmental Engineering graduate student

Mingjun Jiang

1st year Master in Environmental Engineering



UNIVERSITY OF
LIVERPOOL

Stanford

School of Engineering &
Doerr School of Sustainability

Civil & Environmental Engineering

- Bachelor's degree in Biological Sciences from Xi'an Jiaotong-Liverpool University, 2024
- Research experience across multiple Chinese institutions:
 - Microbial community construction mechanism within active sludge of photobioreactor
 - Bioleaching of metals from E-wastes/biotreatment of industrial sewage
 - Functional genomic features of the gut microbiome in colorectal cancer
 - Protein engineering to control cell signalling
- Master student in Environmental Engineering at Stanford University, 2026
 - Wastewater AMR surveillance, sustainable biotechnology, synthetic biology

Background

Antimicrobial resistance (AMR) is one of the top global public health and development threats.

It is estimated that bacterial AMR was directly responsible for 1.27 million global deaths and contributed to 4.95 million deaths in 2019
-WHO

- **Antimicrobials**—including antibiotics, antivirals, antifungals, and antiparasitics—are used to prevent and treat infections in humans, animals, and plants.
- **Antimicrobial resistance (AMR)** arises when microorganisms evolve mechanisms to survive exposure to antimicrobial agents.
- As resistance increases, standard treatments become ineffective, leading to persistent infections, higher morbidity and mortality, and greater risk of transmission.

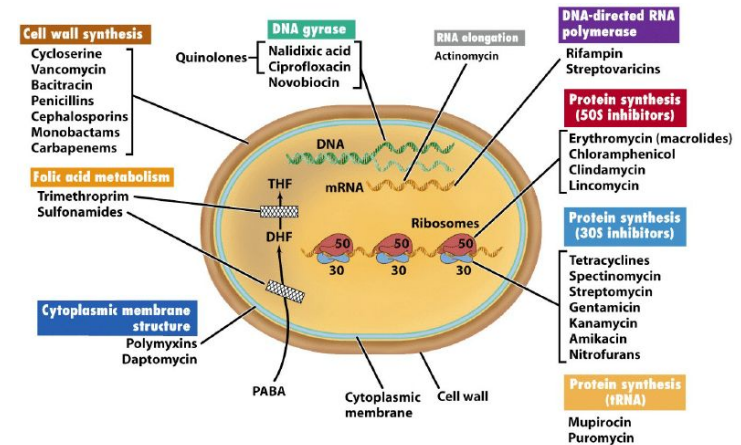


Figure 20-14 Brock Biology of Microorganisms 11/e
© 2006 Pearson Prentice Hall, Inc.

9 types of antibiotic targets

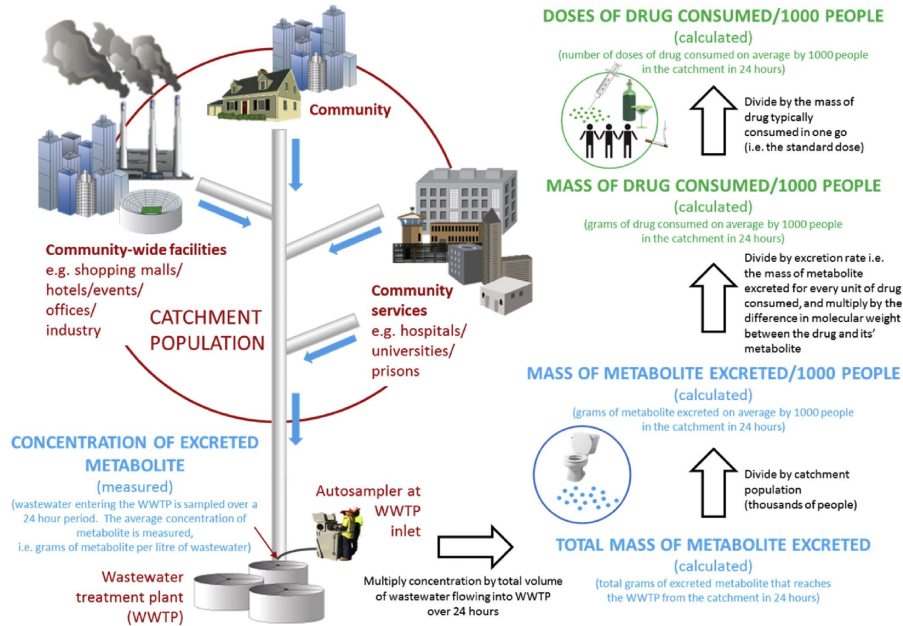
Background

Why WBE for AMR surveillance?

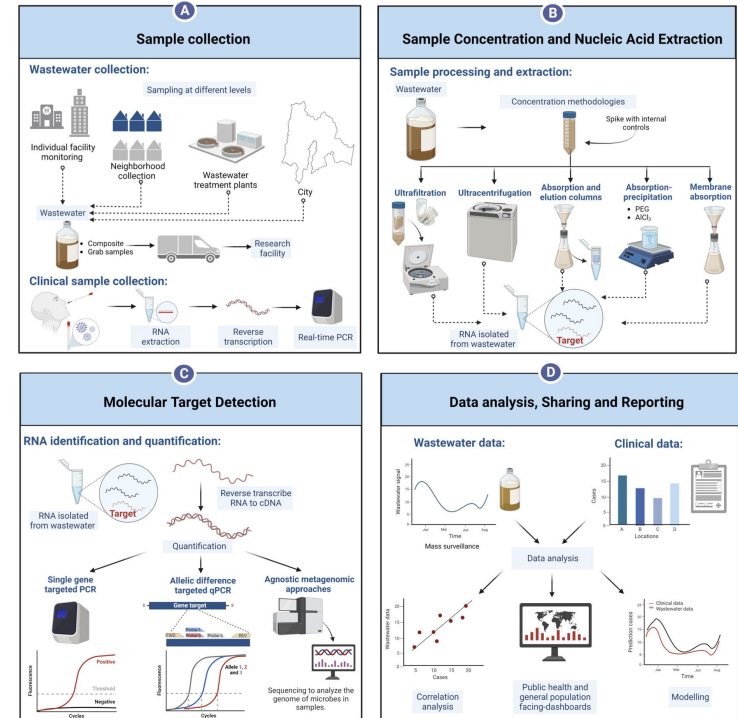
Wastewater-based epidemiology (WBE)

Analyze wastewater to determine the consumption of or exposure to chemicals or pathogens in a population.

Typical workflow of WBE studies



Typical workflow of WBE for Virus & AMR



Choi et al. (2018); Gruden C.R. et al. (2023)

Background

Why WBE for AMR surveillance?

WBE advantage

- **Captures community-wide signals**
Integrates data from both symptomatic and asymptomatic individuals, reflects real-world antibiotic usage and resistance gene transmission
- **Supports early detection and trend monitoring**
Detects emergence or increase of AMR signals before clinical systems respond.
- **Offers spatial and temporal resolution**
Reveals geographic and temporal differences in ARG abundance across neighborhoods, cities, or countries.
- **Non-invasive, cost-effective, scalable**
Avoids individual testing, allowing for low-cost and high-frequency sampling, even in low-resource settings.
- Hospital wastewater serves as a hotspot for ARGs

Autosampler device for hospital wastewater



Recent Studies

Hospital WBE PCR method summary

<i>Authors</i>	<i>PCR method</i>	<i>ARG detection</i>	<i>Quantification</i>	<i>Microbial Typing</i>	<i>Pathogenicity Context</i>	<i>Isolate Genotyping (ARG/Virulence)</i>
Babler et al. (2022)	qPCR , isolate qPCR	✓	✓			✓
Bian et al. (2023), Majeed et al. (2021)	qPCR, HT-qPCR (real time PCR)	✓	✓			
Caplin et al.(2008)	target PCR, isolate PCR, MLST PCR	✓		✓		
Daoud et al.(2018), Drieux et al.(2016), Pignato et al.(2010)	target PCR	✓				
Davidova-Gerzova et al.(2024), Davidova-Gerzova et al.(2023)	Isolate PCR and MLST	✓		✓		✓
Duran-Bedolla et al.(2024), Galvin et al.(2010), Hoffmann et al.(2023), Mbangi et al.(2023), Oravcova et al.(2017), Wu et al.(2023), Xie et al. (2023)	Isolate PCR	✓				✓
Flach et al.(2021)	Isolate PCR, qPCR	✓	✓			✓
Kovacic et al.(2017), Jakobsen et al.(2008)	Isolate and multiplex PCR	✓			✓	✓
Meir-Gruber et al.(2016)	Real RT-PCR	✓			✓	
Rolbiecki et al.(2024)	Isolate and ERIC-PCR			✓		✓
Zhong, Zhang (2013)	Isolate PCR and inverse PCR	✓			✓	✓

Recent Studies

How AMR is studied by WBE

Untargeted approach Metagenomics

Sequences all DNA in the sample, annotation ARGs, hosts, MGEs, pathogens

- Common method: Illumina (shotgun), Oxford Nanopore (Long-read sequencing)

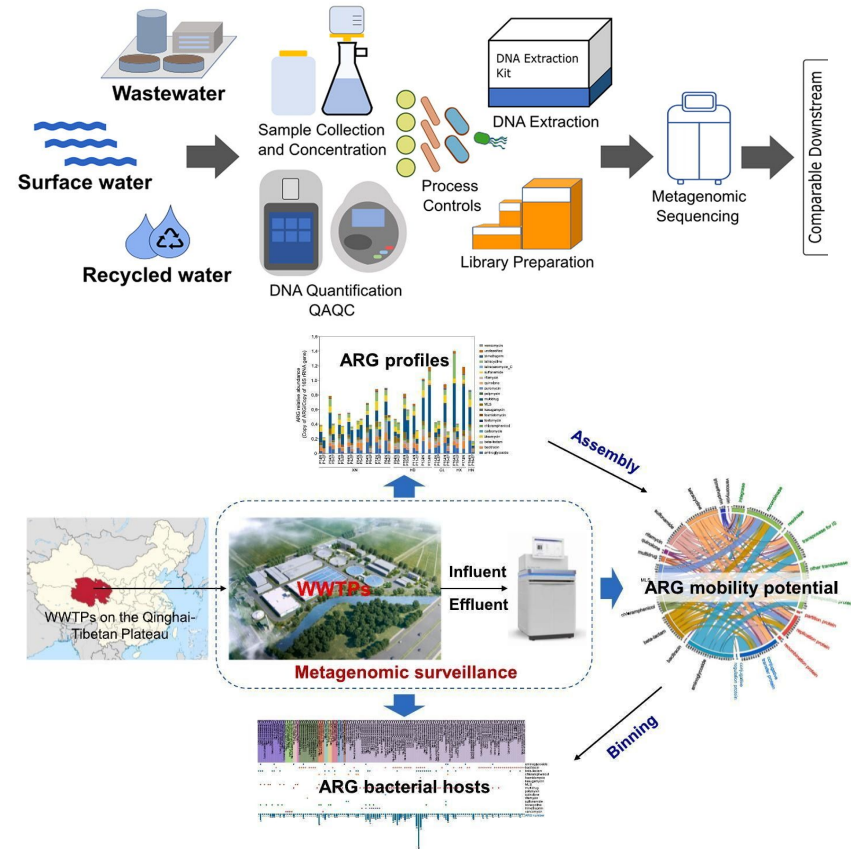
Strengths:

- Captures known and novel ARGs, host attribution, microbial community profiling, horizontal gene transfer

Limitations:

- Rely heavily on sequencing data quality and advanced bioinformatic tools (particularly for genomic assembly and binning), lower sensitivity for rare genes, complexity

Workflow of WBE metagenomic analysis



Shi et al. (2023); Davis. et al. (2023)

Recent Studies

Key findings from current WBE AMR research

- **ARGs are widespread but geographically patterned**

ARGs such as ***sul1***, ***tetA***, ***blaTEM***, ***qnrS***, and ***mecA***, are present in worldwide urban wastewater. Their differences in abundance may reflect local antibiotic usage and healthcare practices. (Hendriksen et al., 2019; Karkman et al., 2020)

- **Hospital WBE may reflect clinical trends**

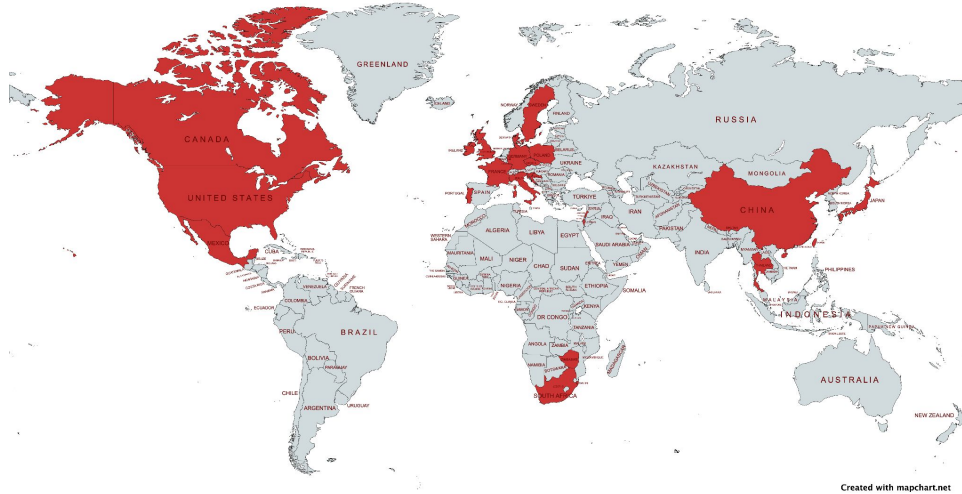
ARG profiles in hospital wastewater are similar to resistance patterns in hospitals. WBE may detect emerging infection trends earlier than clinical reports (Gaze et al., 2021; Hutinel et al., 2019).

- **Co-selection and ARG-MGE Linkage**

Mobile genetic elements, including integrons ***intl1***, plasmid markers ***repA***, heavy metal resistance genes ***copA*** always co-occurrence with ARGs, which may serve as potential point for horizontal ARG transfer study (Di Cesare et al., 2016)

Research

A map of countries from selected articles



Aim:

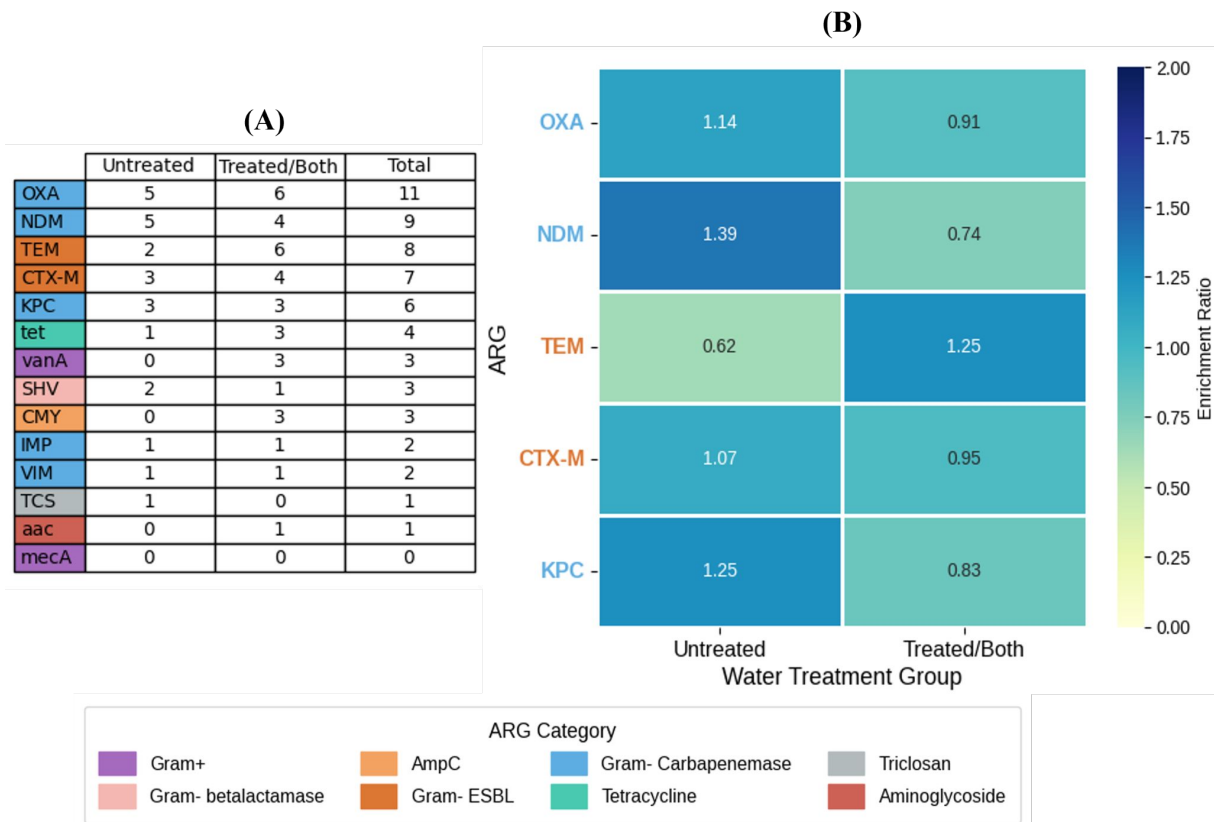
Assess the concordance between AMR levels in wastewater and hospital infection

Results:

- 37 articles from 23 countries
- China, United Kingdom and Czech Republic as most frequently reported
- 78% of the included studies demonstrated concordance between AMR patterns in wastewater and clinical samples.

Research

ARG reporting by wastewater treatment method



Research

ARG reporting by country-income-level

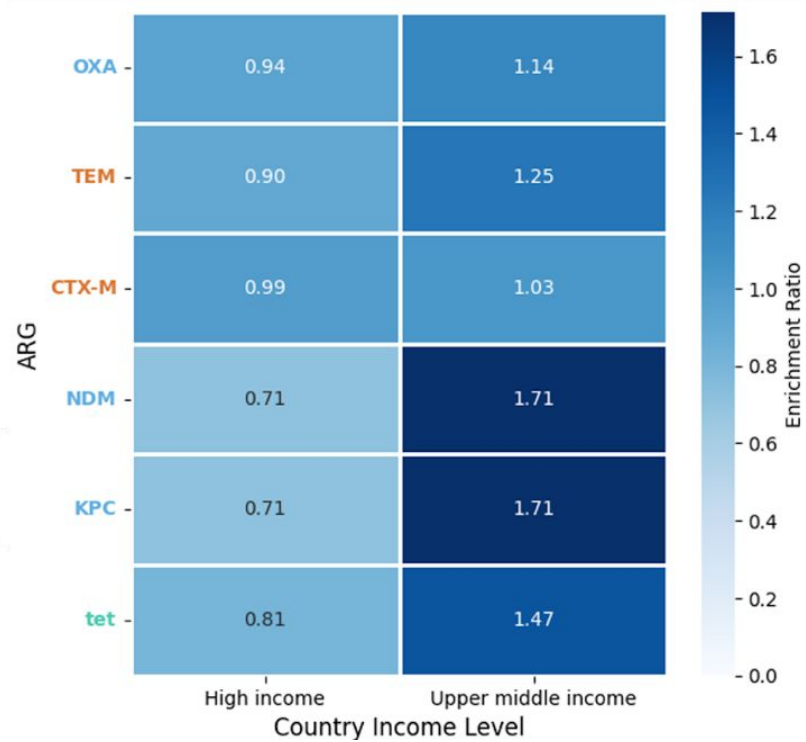
(A)

	High income	Upper middle income	Lower middle income	Total
OXA	8	4	1	13
TEM	7	4	1	12
NDM	5	5	1	11
CTX-M	7	3	1	11
KPC	4	4	0	8
tet	4	3	0	7
VIM	3	1	0	4
SHV	2	1	1	4
vanA	4	0	0	4
CMY	2	1	0	3
aac	2	1	0	3
IMP	3	0	0	3
mecA	1	0	0	1
TCS	1	0	0	1

ARG Category



(B)



Future research framework



Collect hospital
wastewater

Biweekly collect
hospital wastewater and
extract the DNA



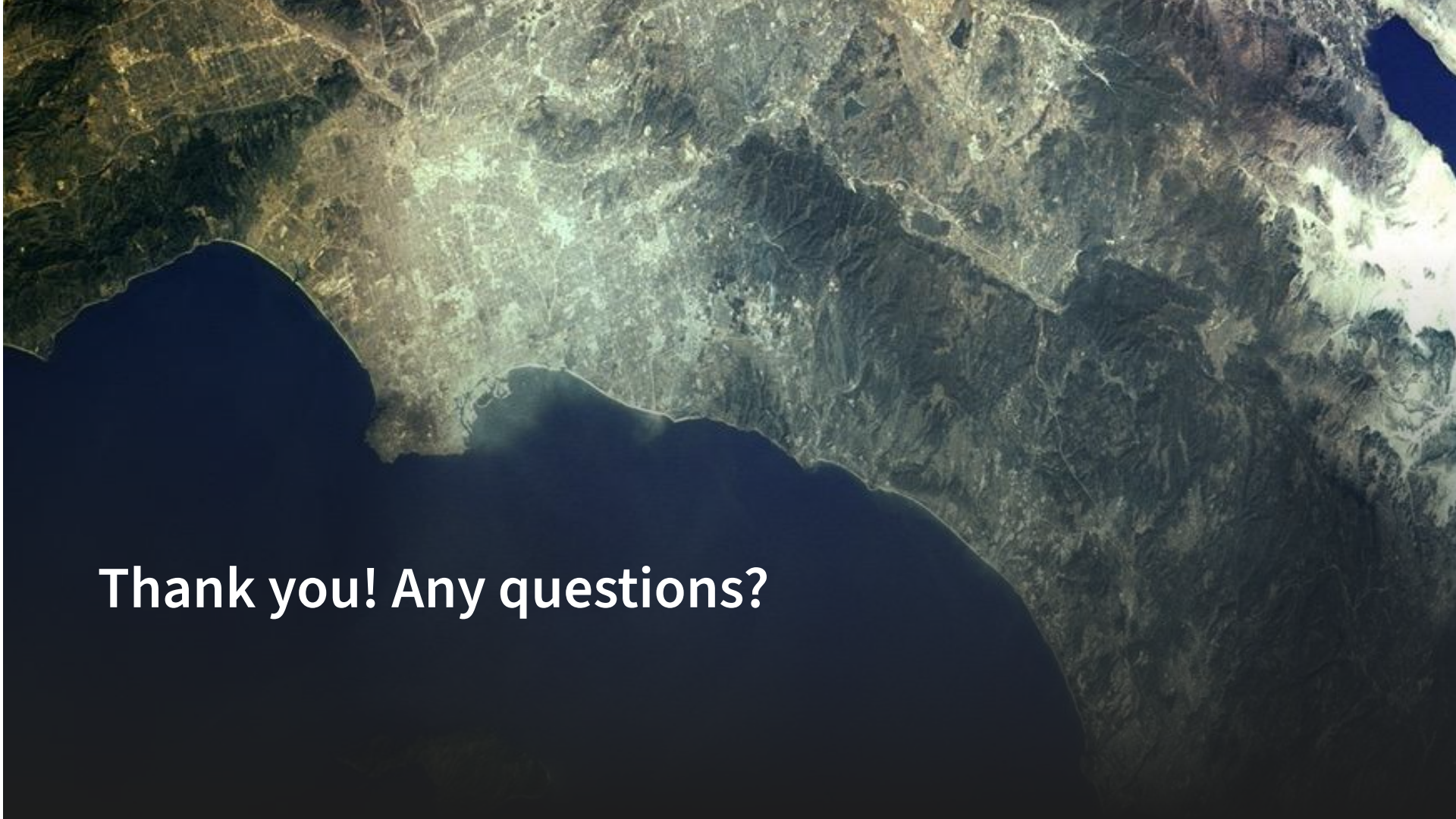
Metagenomic &
PCR target



WBE findings align
with clinical data



Build ARG-host
network, detect ARG
transfer patterns,
identify biomarkers

A satellite image showing a rugged coastline. The land is covered in dense, dark green vegetation, with some lighter brown patches indicating cleared areas or different types of terrain. The coastline is irregular, with several small bays and peninsulas. The sea is a deep, dark blue, contrasting sharply with the land. The overall scene is a high-contrast, aerial view of a natural landscape.

Thank you! Any questions?